

Sri Lanka Institute of Information Technology



BSc (Hons) in Information Technology Specializing in
Data Science

IT3021 – Data Warehousing and Business Intelligence

Assignment 2

IT Number : IT23744516

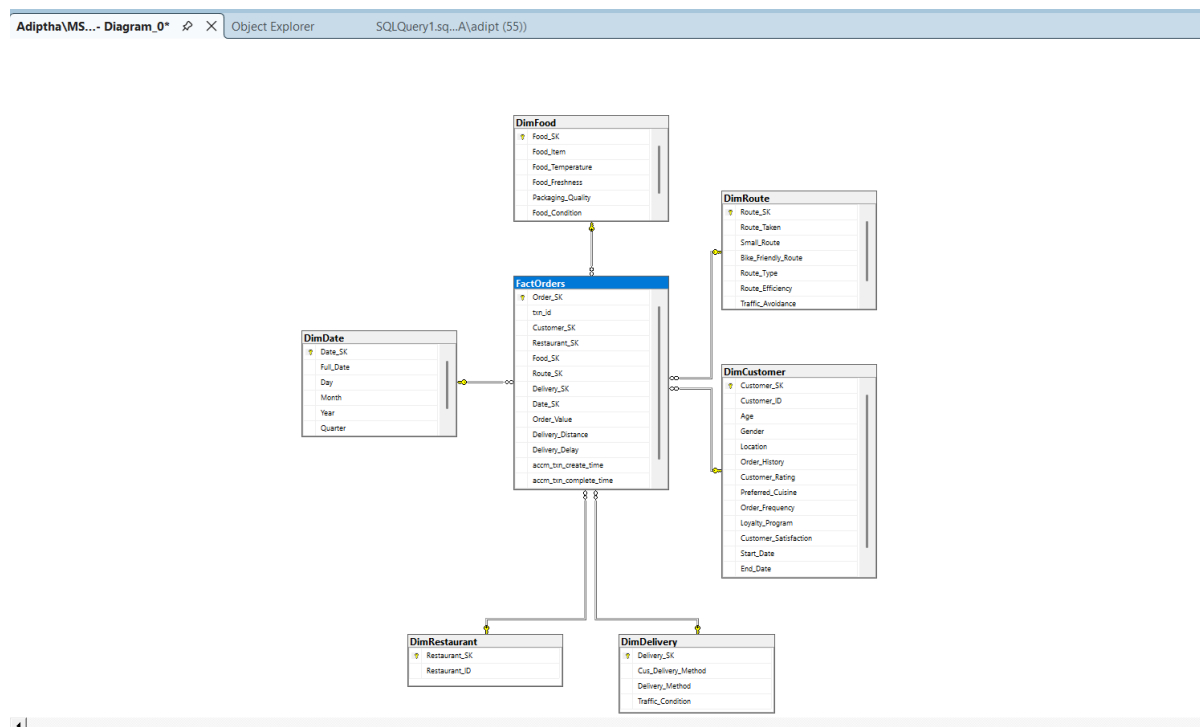
Name : K.B.T.A Dissanayake

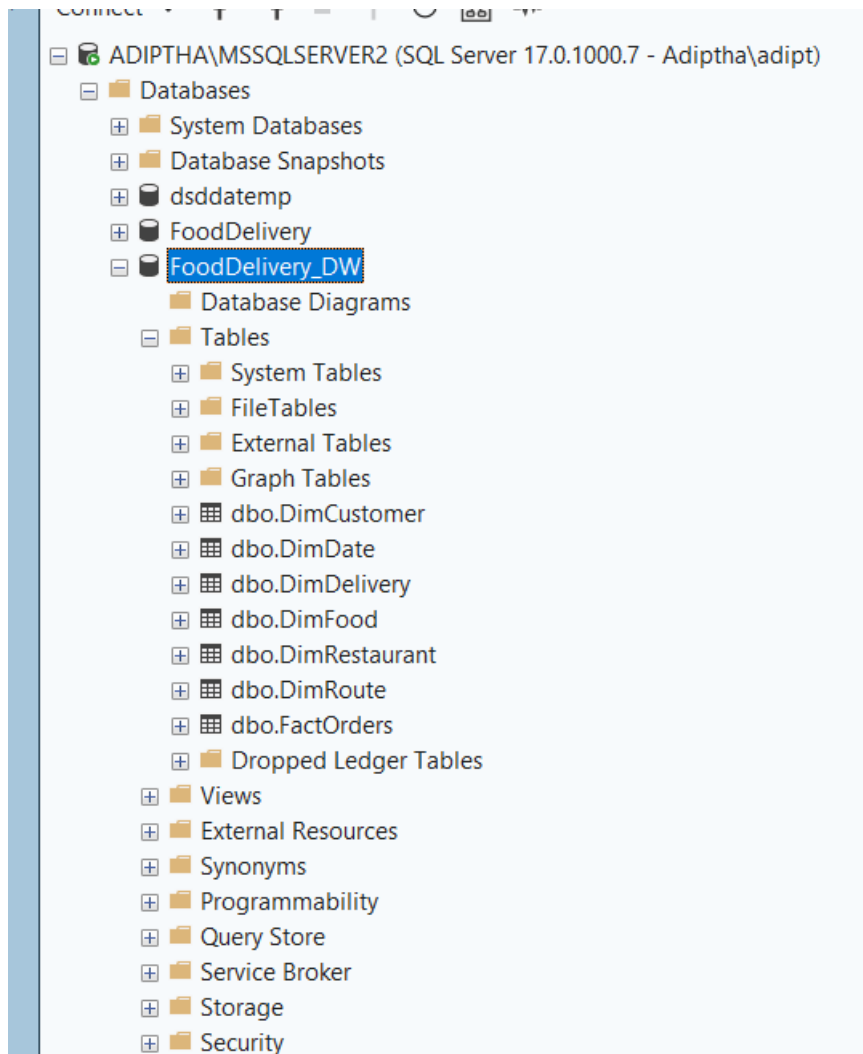
Task 1: Data source for the Assignment 2

The data source used for this assignment is a **Data Warehouse developed in Assignment 1**, based on an **Online Food Delivery Order Management System**.

The data warehouse was designed to store and analyze transactional data related to food delivery operations, including customer details, delivery information, food attributes, routes, and order transactions.

The system follows a **star schema design**, where a central fact table is connected to multiple dimension tables to support analytical queries and reporting.





FactOrders

The FactOrders table stores the measurable transactional data of the system.

Key attributes include:

- Order_Value
- Delivery_Distance
- Delivery_Delay
- txn_process_time_hours
- accm_txn_create_time
- accm_txn_complete_time

Foreign keys:

- Customer_SK
- Date_SK
- Delivery_SK
- Food_SK
- Route_SK

- Restaurant_SK

This table is used to perform aggregations such as total order value, order count, and delivery analysis.

4. Dimension Tables

The following dimension tables provide descriptive information:

DimCustomer

Contains customer-related details such as:

- Customer_SK
- Gender
- Location
- Loyalty_Program
- Preferred_Cuisine
- Customer_Rating

DimDate

Provides time-based hierarchy for analysis:

- Date_SK
- Year
- Quarter
- Month
- Day

This dimension supports time-based analysis such as drill-down operations.

DimDelivery

Stores delivery-related attributes:

- Delivery_SK
- Delivery_Method (Bike, Car, Walk)
- Traffic_Condition

DimFood

Contains food-related attributes:

- Food_SK
- Food_Item
- Food_Condition
- Food_Freshness
- Packaging_Quality

DimRoute

Provides route and navigation details:

- Route_SK
- Route_Taken

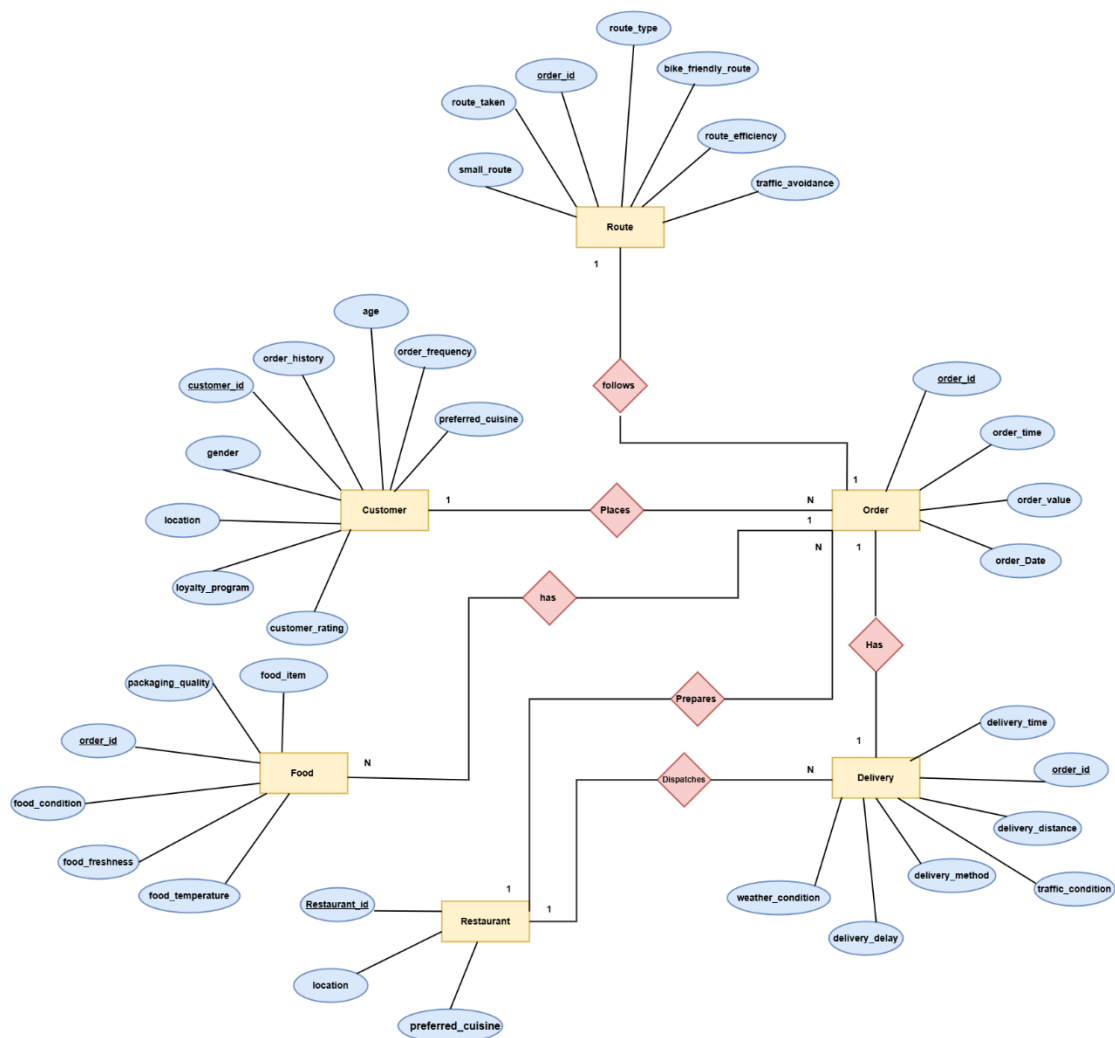
- Route_Type
- Traffic_Avoidance
- Route_Efficiency

DimRestaurant

Contains restaurant information:

- Restaurant_SK
- Restaurant_ID

ER Diagram:



Task 2: SSAS Cube implementation

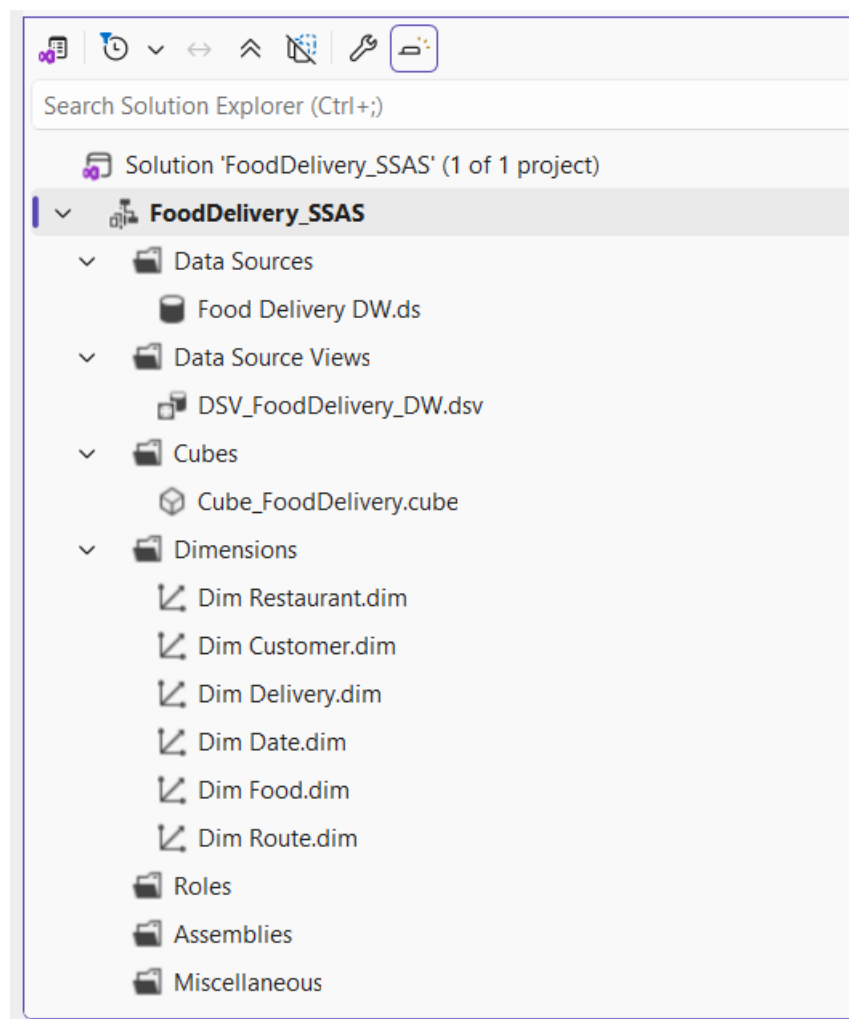
The SSAS cube was developed using SQL Server Analysis Services (SSAS) to enable multidimensional analysis of the data warehouse. The cube was created using the **FoodDelivery_DW** database as the data source and designed to support OLAP operations such as roll-up, drill-down, slicing, and dicing.

A new **Analysis Services Multidimensional Project** was created using SQL Server Data Tools (SSDT).

The following steps were performed:

- Created a new SSAS project
- Configured deployment settings
- Connected to SQL Server Analysis Services instance

FoodDelivery_SSAS.Slnx

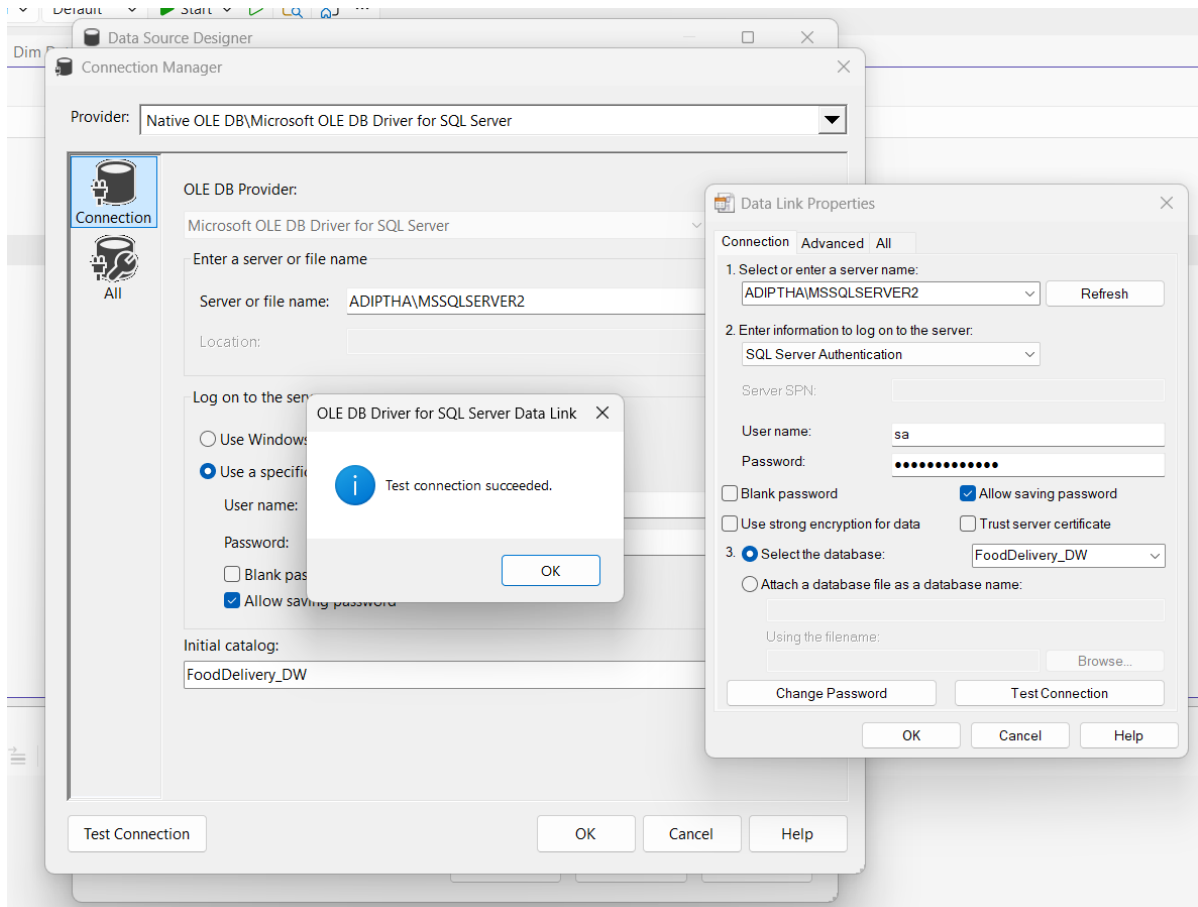


2.1 Data Source Configuration

A data source was created to connect the cube to the data warehouse.

- Data Source: FoodDelivery_DW
- Connection type: SQL Server

This allows SSAS to retrieve data from the data warehouse.



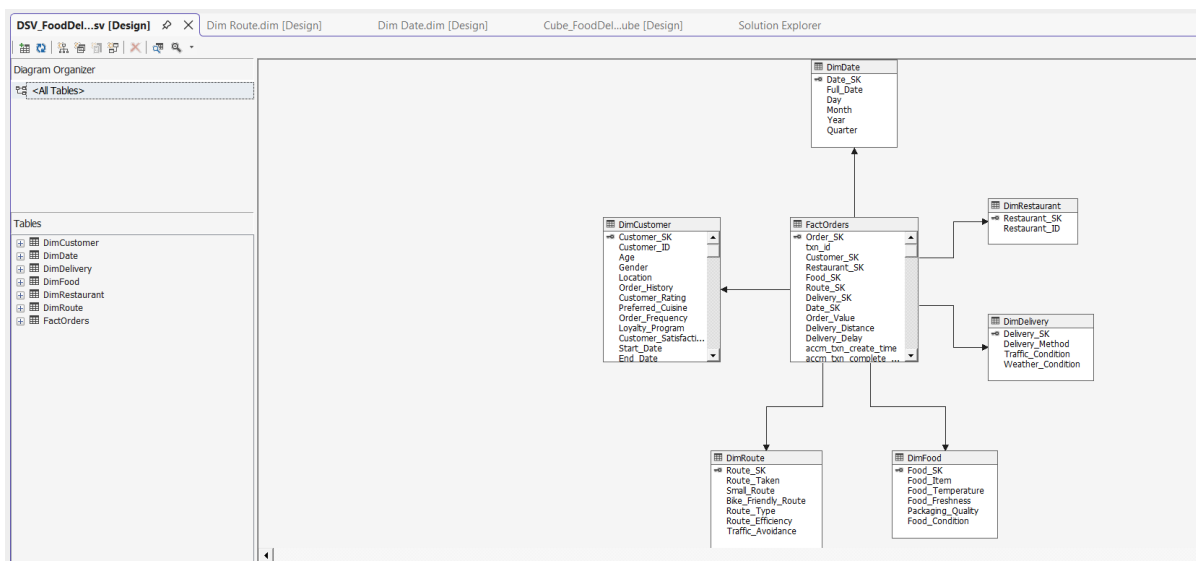
2.2 Data Source View (DSV)

A Data Source View was created to define the tables used in the cube.

The following tables were included:

- FactOrders
- DimCustomer
- DimDate
- DimDelivery
- DimFood
- DimRoute
- DimRestaurant

Relationships between tables were automatically detected using foreign keys.



2.3 Cube Creation

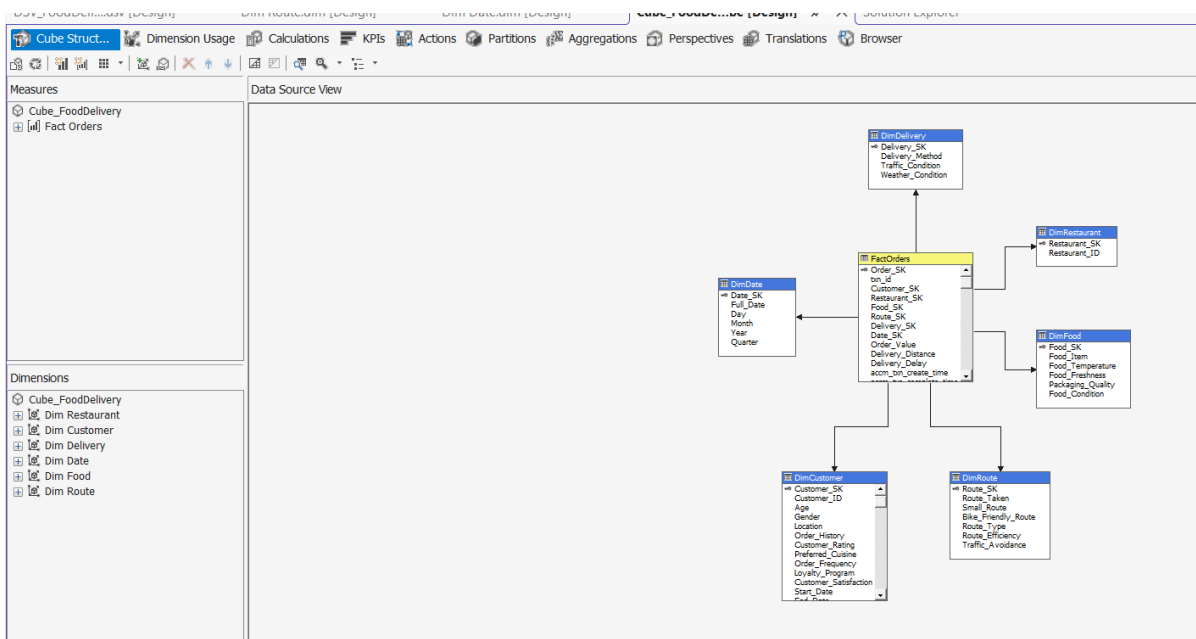
The cube was created using the Cube Wizard.

Measures:

The following measures were selected from the FactOrders table:

- Order Value
- Delivery Distance
- Delivery Delay
- txn_process_time_hours

These measures are used for aggregation and analysis.



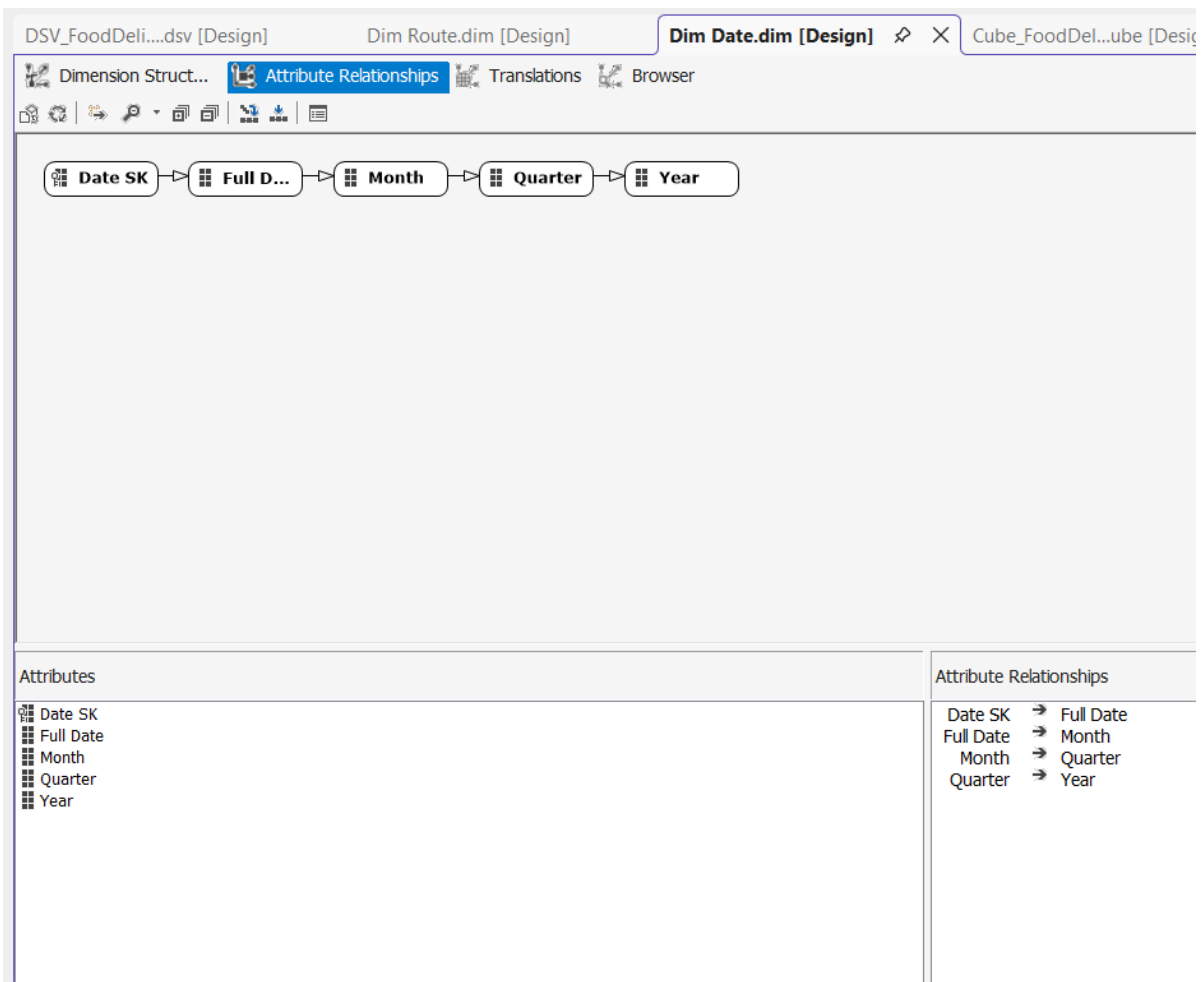
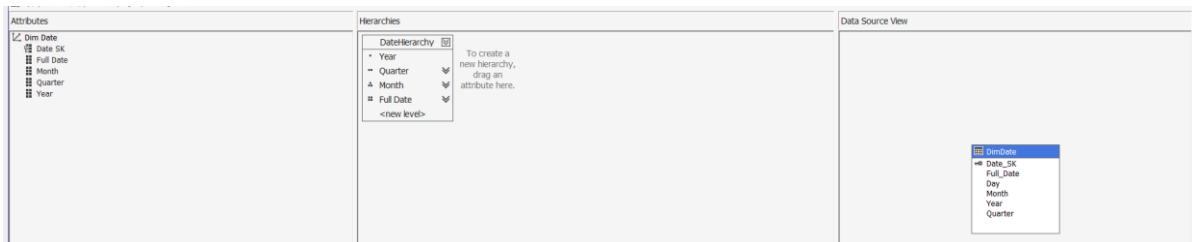
2.4 Hierarchy Implementation

A hierarchies were implemented in from the Dimensions:

Example:

Year → Quarter → Month → Day

This hierarchy allows users to perform drill-down and roll-up operations.

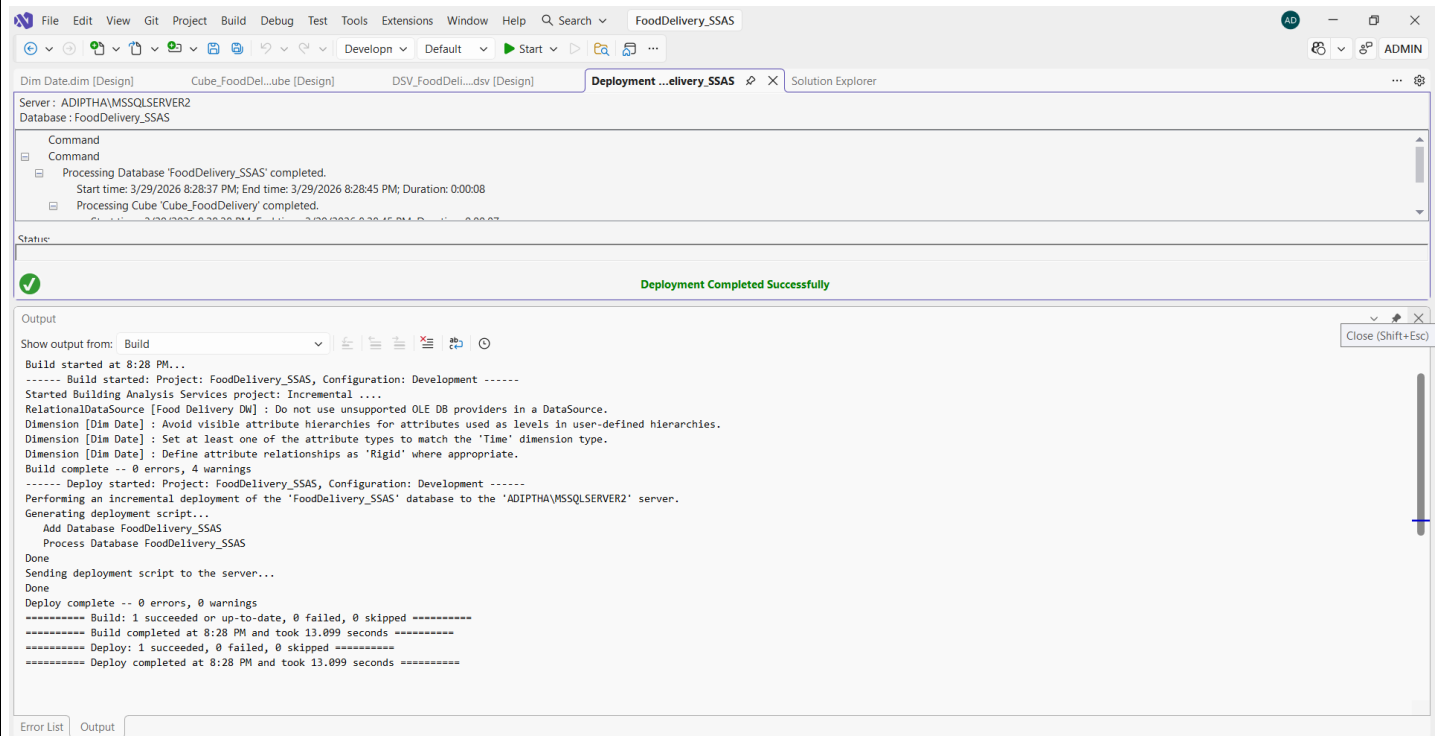


2.5 Cube Deployment and Processing

The cube was deployed to the SSAS server.

After deployment:

- The cube was processed
- Data was loaded into the cube
- Cube was tested using the browser tab



Task 3: Demonstration of OLAP operations

An Excel workbook was connected to the SSAS cube to perform OLAP operations using Pivot Tables and Pivot Charts.

Excel provides powerful analytical capabilities to interact with multidimensional data, allowing users to perform operations such as roll-up, drill-down, slice, dice, and pivot.

Steps I followed,

3.1 Connecting Excel to SSAS Cube

The following steps were followed to connect Excel to the SSAS cube:

1. Opened **Microsoft Excel**
2. Navigated to the **Data tab**
3. Selected: Get Data → From Database → From Analysis Services
4. Entered the **SSAS server name**
5. Selected the deployed cube (**FoodDelivery Cube**)
6. Clicked **Finish**
7. Selected: PivotTable Report
8. Clicked **OK**

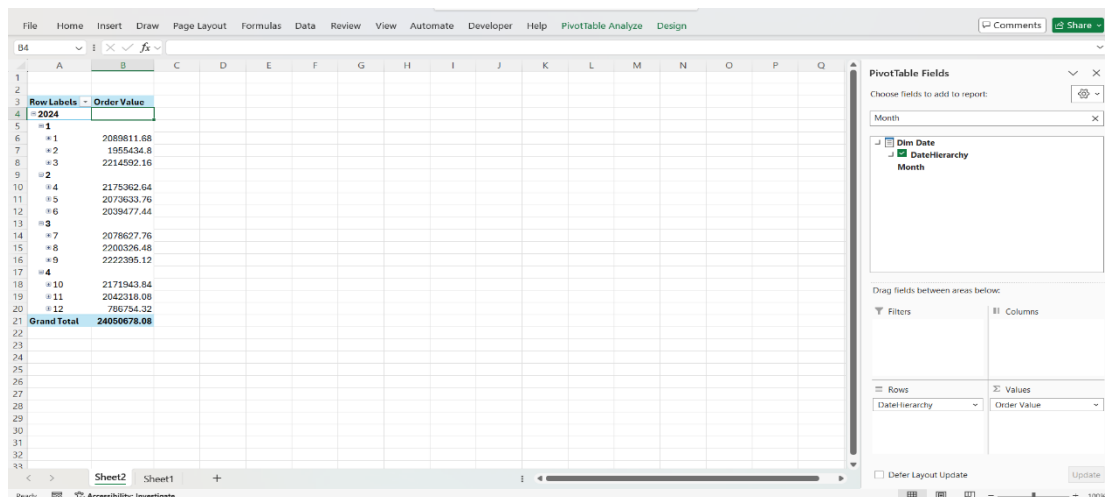
3.2 Creating PivotTable for Analysis

A PivotTable was created to analyze cube data.

The fields were arranged as follows:

- **Rows:** Date Hierarchy (Year → Month)
- **Values:** Total Order Value

This structure enabled multidimensional analysis of order data across time and customer segments.



3.3 Roll-Up Operation

Roll-up aggregates data to a higher level in the hierarchy.

Implementation:

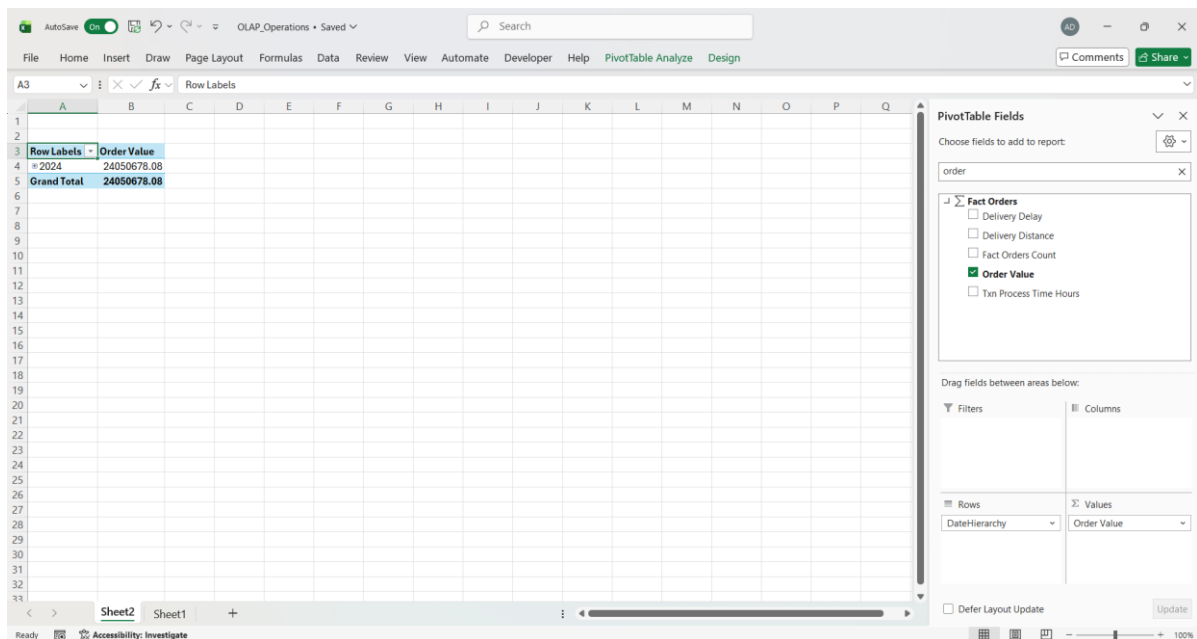
- The PivotTable was set to display only the **Year level**
- Monthly data was aggregated into yearly totals

Result:

- Users can view total order value per year

Explanation:

This operation provides a summarized view of the dataset.



3.4 Drill-Down Operation

Drill-down allows users to move from summary data to detailed data.

Implementation:

- Expanded the Year level to display **Month-level data**
- Used the “+” icon in the PivotTable

Result:

- Users can analyze monthly variations in order value

Explanation:

This operation enables detailed analysis of trends over time.

Row Labels	Order Value
= 2024	
#1	
#1	2089811.68
#2	1955434.8
#3	2214592.16
#2	
#4	2175362.64
#5	2073633.76
#6	2039477.44
#3	
#7	2078627.76
#8	2200326.48
#9	2222395.12
#4	
#10	2171943.84
#11	2042318.08
#12	786754.32
Grand Total	24050678.08

3.5 Slice Operation

Slice filters data based on a single dimension.

Implementation:

- Applied a filter using: Food Condition
- Selected a single value (e.g., **good**)

Result:

- PivotTable displays only data related to that delivery method

Explanation:

Slice reduces the dataset to a specific subset based on one condition.

Food Condition	Order Value
Good	
#4	
#10	1820618.8
#11	1658169.04
#12	625239.12
Grand Total	19744719.38

3.6 Dice

Operation

Dice filters data using multiple dimensions simultaneously.

Implementation:

- Applied filters using:
 - Delivery Method
 - Route Taken
 - Food Condition

Result:

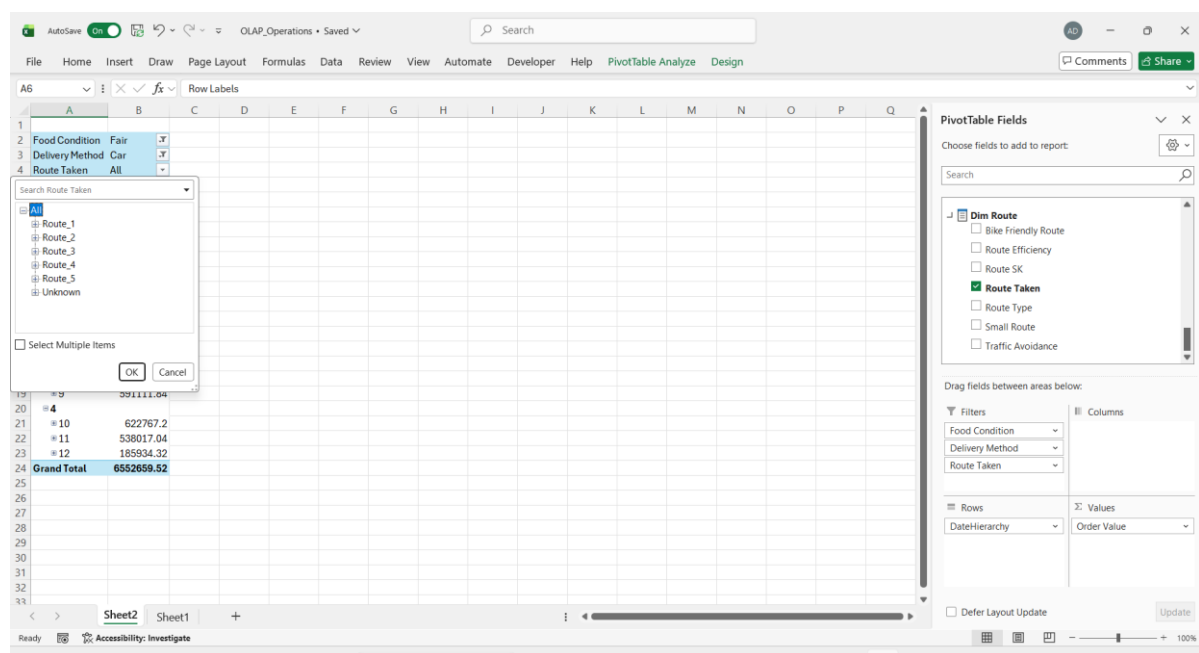
- Data is filtered based on a combination of conditions

Example:

- Delivery Method = Car
- Route Taken = Route_1
- Food Condition = Good

Explanation:

Dice enables more complex and precise data analysis.



3.7 Pivot Operation

Pivot rotates the data view to provide a different perspective.

Implementation:

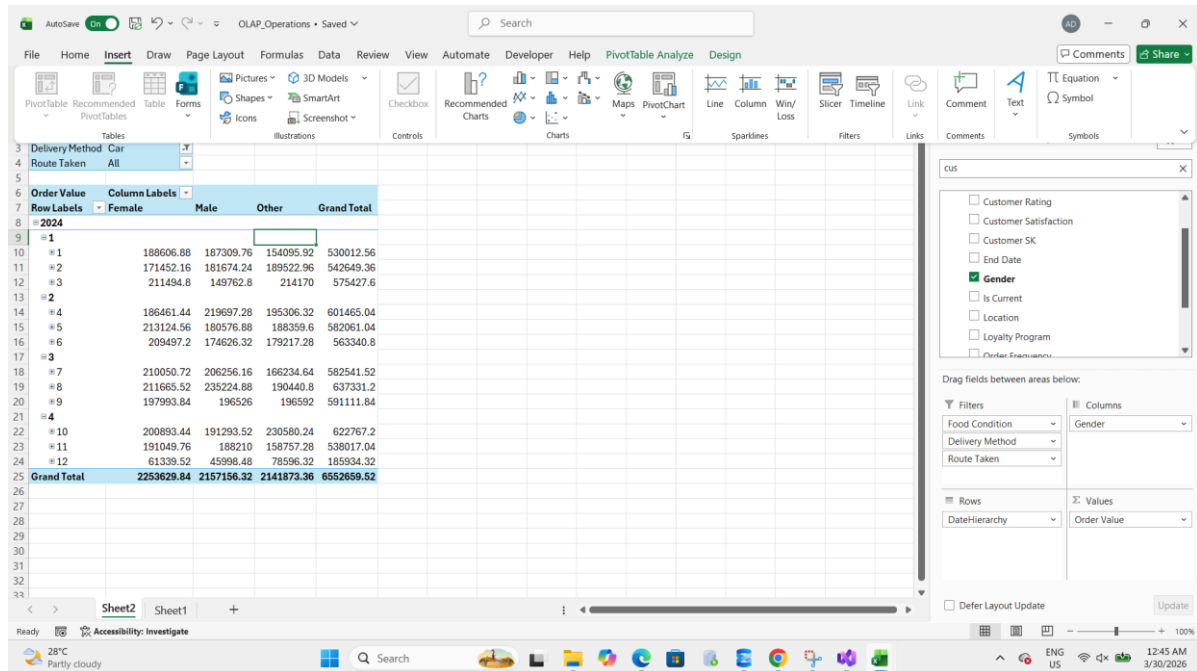
- Moved **Gender** from Columns to Rows
- Rearranged fields within the PivotTable

Result:

- Data presentation changes, enabling comparison from a different angle

Explanation:

Pivot allows flexible reorganization of dimensions for better insights.



3.8 Pivot Chart Visualization

A Pivot Chart was created from the PivotTable to visualize the data.

Steps:

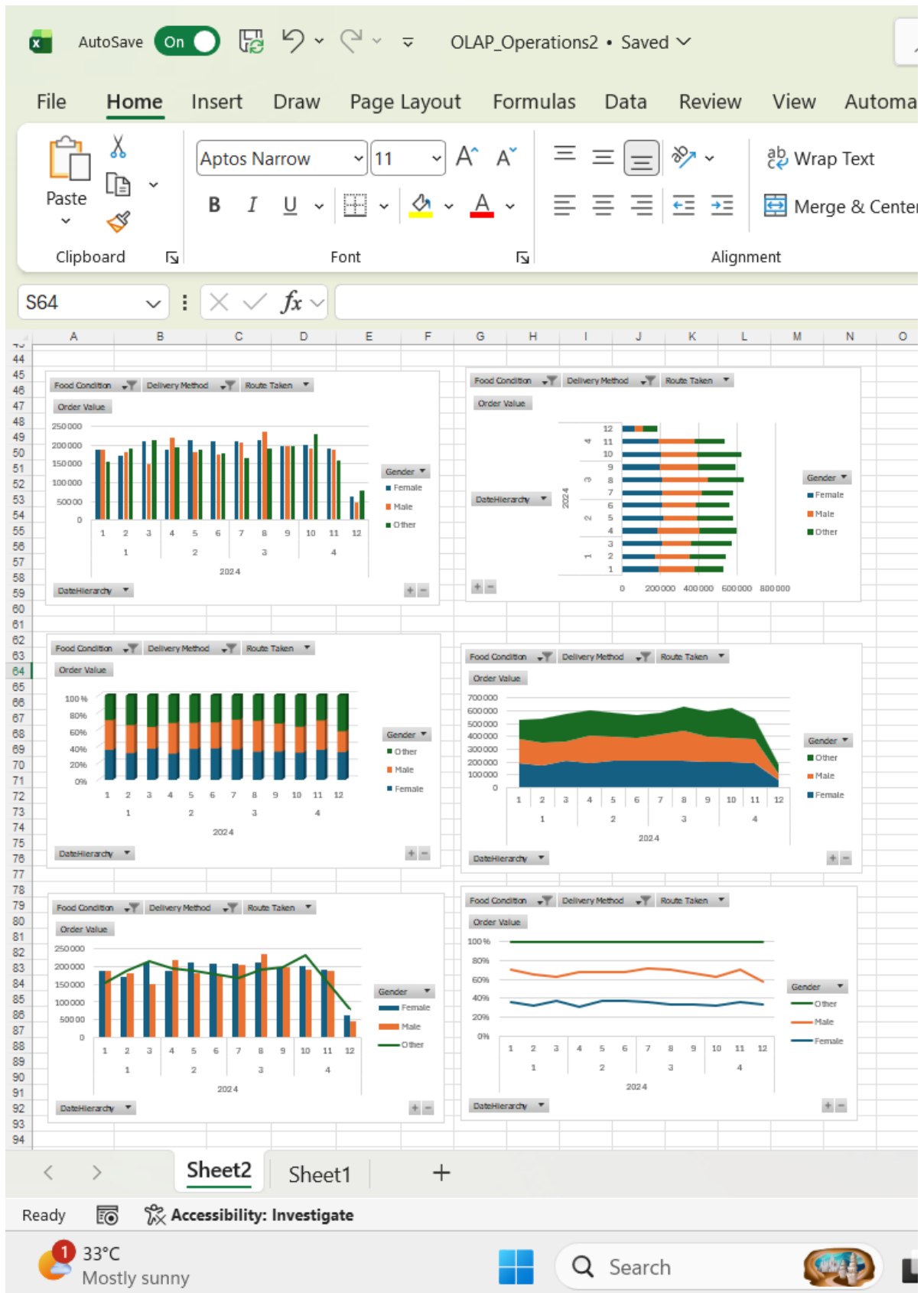
1. Selected the PivotTable
2. Navigated to: Insert → PivotChart
3. Selected a chart type (e.g., Column Chart)

Result:

- Visual representation of order value trends

Explanation:

Pivot charts improve readability and support better decision-making.



Task 4: PowerBI Reports

Power BI Desktop was used to develop interactive reports based on the data warehouse. The reports were designed to provide insights using visualizations, filters, and advanced features such as drill-down and drill-through.

The reports were published to **Power BI Service (online)** to enable web-based access and demonstration.

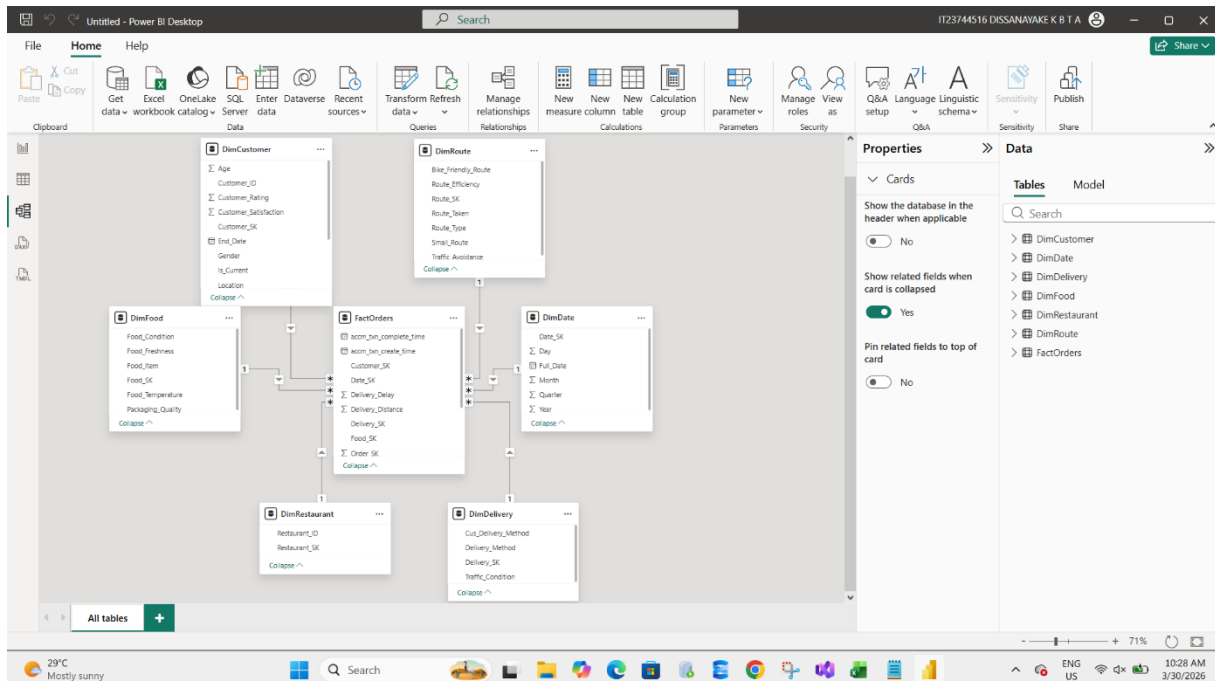
4.1 Data Preparation and Modeling

The following steps were performed:

- Connected Power BI to the SQL Server Data Warehouse
- Imported fact and dimension tables
- Established relationships between tables using surrogate keys
- Verified the star schema model

Tables used:

- FactOrders
- DimCustomer
- DimDate
- DimDelivery
- DimFood
- DimRoute



Measures Created

The following measures were created using DAX:

Total Order Value = SUM(FactOrders[Order_Value])

Order Count = COUNTROWS(FactOrders)

Total Delivery Distance = SUM(FactOrders[Delivery_Distance])

These measures were used across all reports.

The screenshot shows the Microsoft Power BI Desktop interface. The 'Measure tools' ribbon is active, displaying the 'Name' field as 'Average Customer ...', the 'Format' dropdown set to 'General', and the 'Data category' dropdown set to 'Uncategorized'. The 'Structure' pane on the left shows a list of measures: 'Average Customer Rating', 'Total Order Value', 'Total Delivery Distance', and 'Order Count'. The 'Data' pane on the right shows a list of tables: 'DimDelivery', 'DimFood', 'DimRestaurant', 'DimRoute', 'FactOrders', 'accm_txn_complete_time', 'accm_txn_create_time', 'Date Hierarchy', 'Customer_SK', 'Date_SK', 'Delivery_Delay', 'Delivery_Distance', 'Delivery_SK', 'Food_SK', 'Order Value', 'Order_SK', 'Restaurant_SK', 'Route_SK', 'Total Order Value', 'txn_id', and 'txn_process_time_hours'. The main area displays the DAX code for these measures:

```
1 Average Customer Rating = AVERAGE(DimCustomer[Customer_Rating])
2
3 Total Order Value = SUM(FactOrders[Order_Value])
4
5 Total Delivery Distance = SUM(FactOrders[Delivery_Distance])
6
7 Order Count = COUNTROWS(FactOrders)
```

Below the code, a data table is shown with columns: Order_SK, txn_id, Customer_SK, Restaurant_SK, Food_SK, Route_SK, Delivery_SK, Date_SK, Order Value, Delivery_Distance, Delivery_Delay, accm_txn_create_time, and accm_txn_id. The table contains two rows of data:

Order_SK	txn_id	Customer_SK	Restaurant_SK	Food_SK	Route_SK	Delivery_SK	Date_SK	Order Value	Delivery_Distance	Delivery_Delay	accm_txn_create_time	accm_txn_id
49281	ORD002465	7481	96	2344	5946	7	20240721	22.7	2.94	-4		
49282	ORD002465	7481	96	2344	5946	7	20240721	22.7	2.94	-4		

At the bottom, a status bar indicates 'Table: FactOrders (880,000 rows) Column: Average Customer Rating (0 distinct values)'.

The screenshot shows the 'Data' pane in Power BI. It contains a search bar and a list of tables and measures. The tables listed are: 'DimDelivery', 'DimFood', 'DimRestaurant', 'DimRoute', 'FactOrders', 'accm_txn_complete_time', 'accm_txn_create_time', 'Date Hierarchy', 'Customer_SK', 'Date_SK', 'Delivery_Delay', 'Delivery_Distance', 'Delivery_SK', 'Food_SK', 'Order Value', 'Order_SK', 'Restaurant_SK', 'Route_SK', 'Total Order Value', 'txn_id', and 'txn_process_time_hours'. The measures listed are: 'Order Count', 'Total Delivery Distance', and 'Total Order Value'.

4.2 Report 1 – Matrix Visual

A Matrix visual was created to display grouped tabular data.

Implementation:

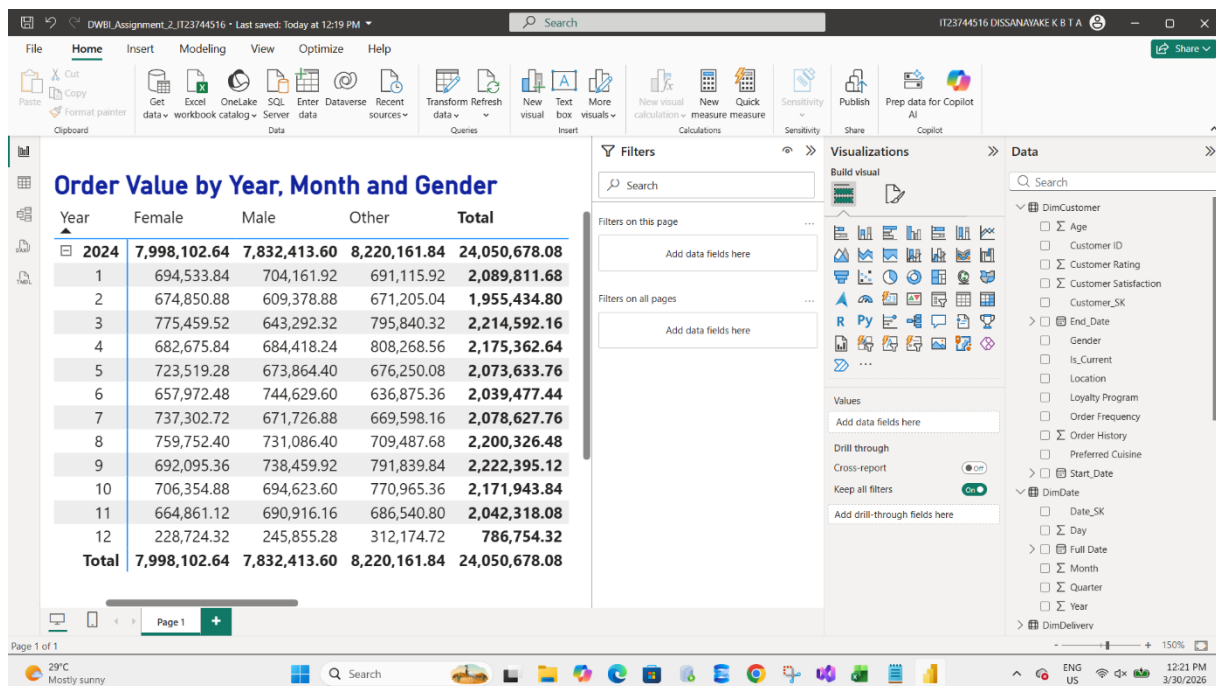
- Rows: Year → Month
- Columns: Gender
- Values: Total Order Value

Purpose:

To analyze order values across time and customer categories.

Outcome:

- Provides hierarchical grouping
- Enables comparison across gender



4.3 Report 2 – Slicers and Cascading Filters

A dashboard was created using multiple slicers and charts.

Slicers Used:

- Gender
- Location

Cascading Filtering:

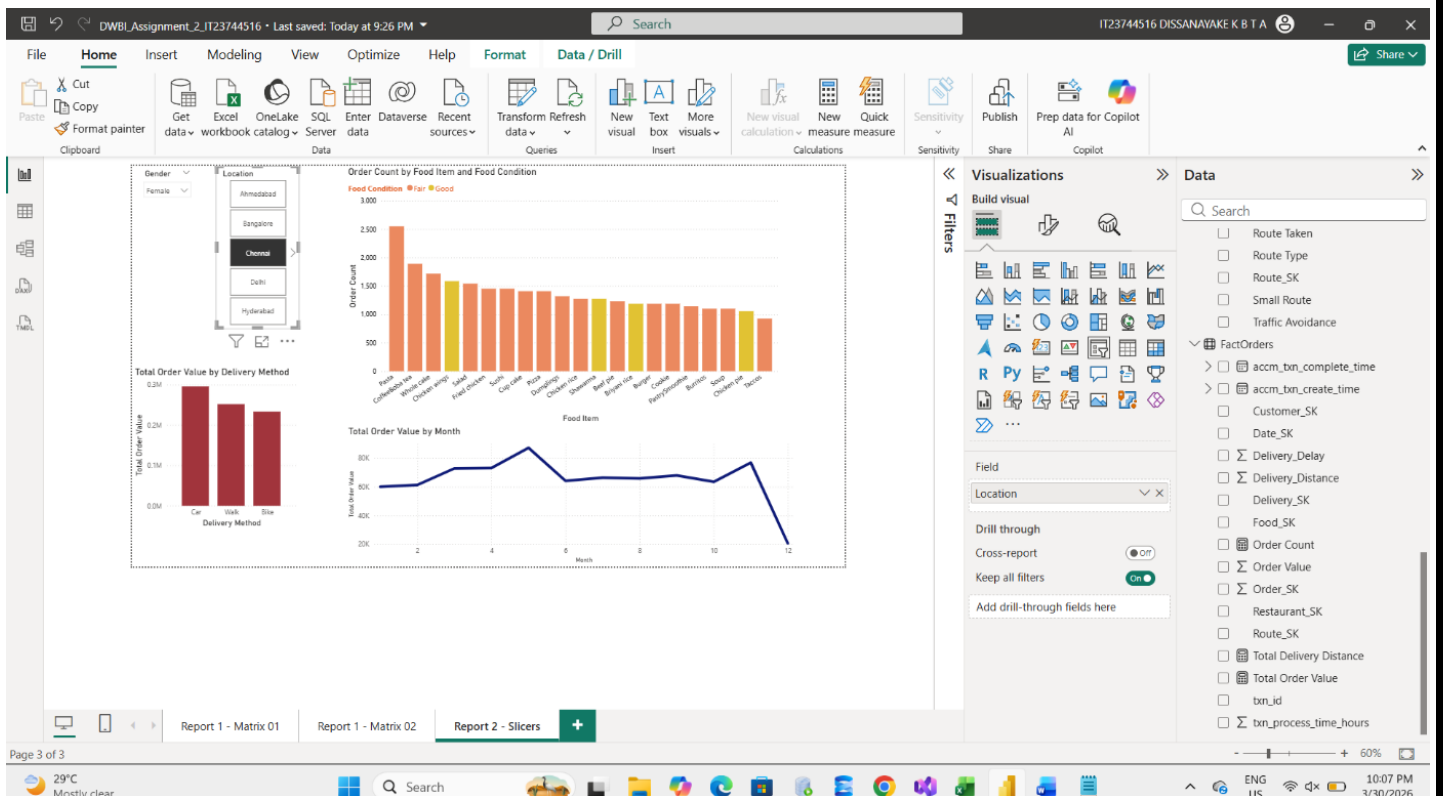
- Selecting Gender dynamically filters Location values

Visuals Used:

- Column chart: Order Value by Delivery Method
- Bar chart: Order Count by Food Item
- Line chart: Monthly Order Value

Purpose:

To allow interactive filtering and data exploration.



4.4 Report 3 – Drill-Down Report

A drill-down report was created using a hierarchical structure.

Implementation:

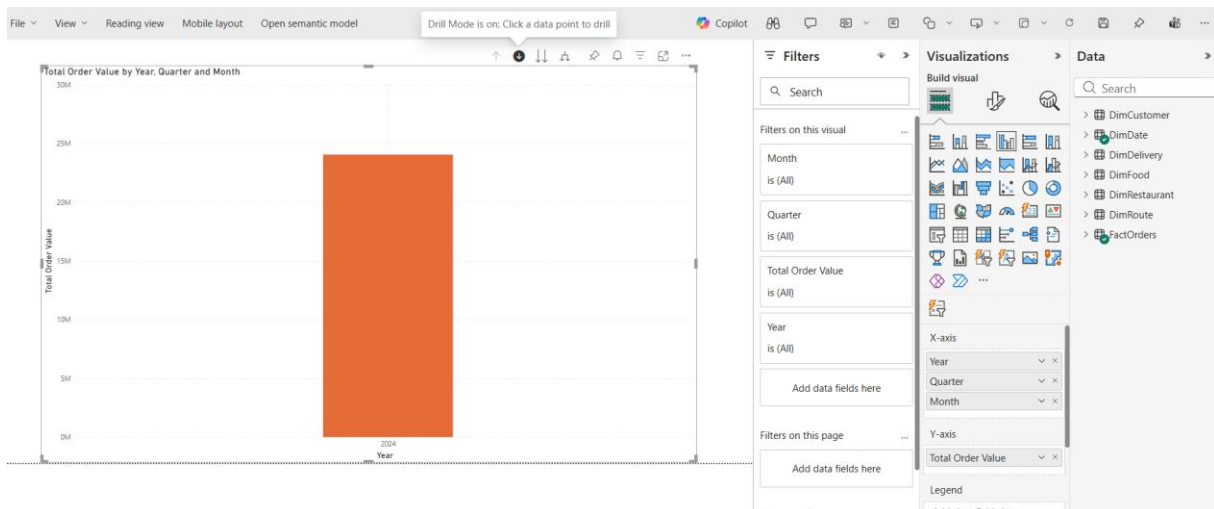
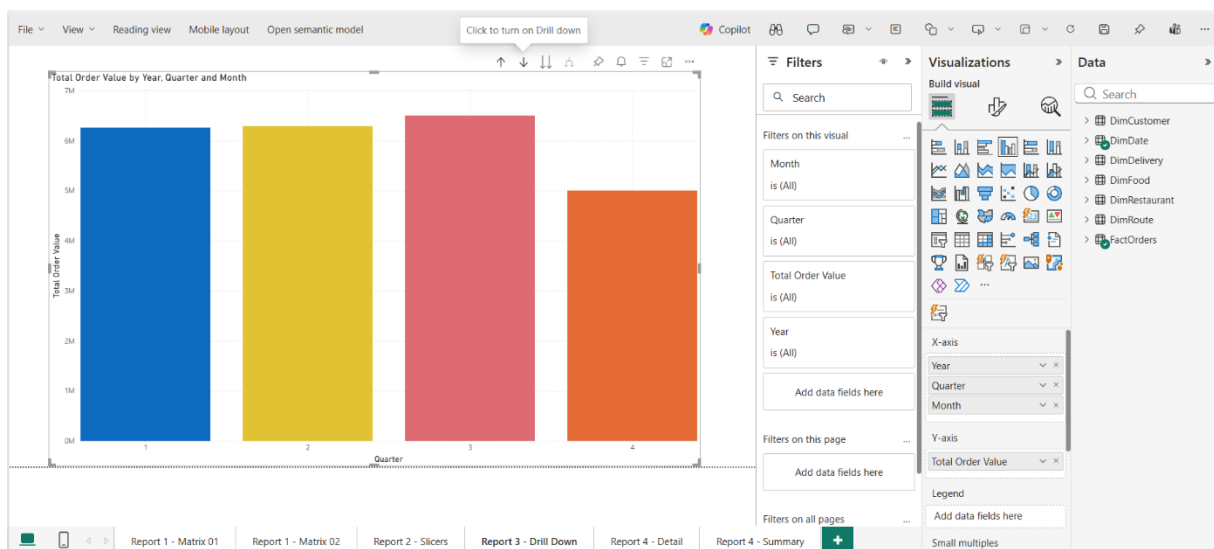
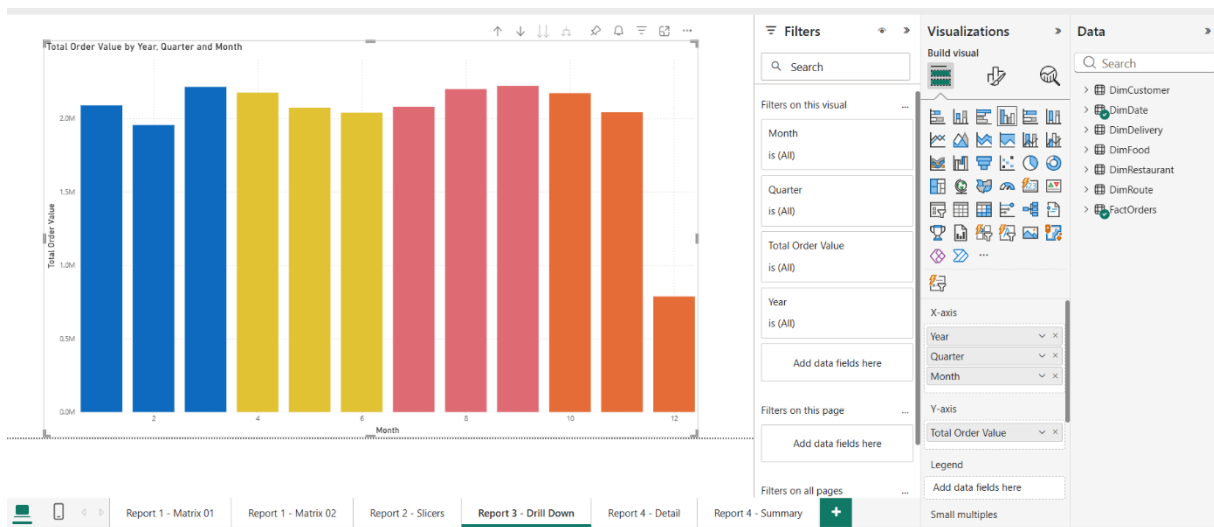
- Axis: Year → Quarter → Month
- Values: Total Order Value

Functionality:

- Users can click Year → drill to Quarter → drill to Month
- Drill-up also supported

Purpose:

To analyze trends at different levels of detail.



4.5 Report 4 – Drill-Through Report

A drill-through report was created to navigate from summary to detailed data.

Summary Page:

- Visual: Order Value by Route Taken (or Location)

Detail Page:

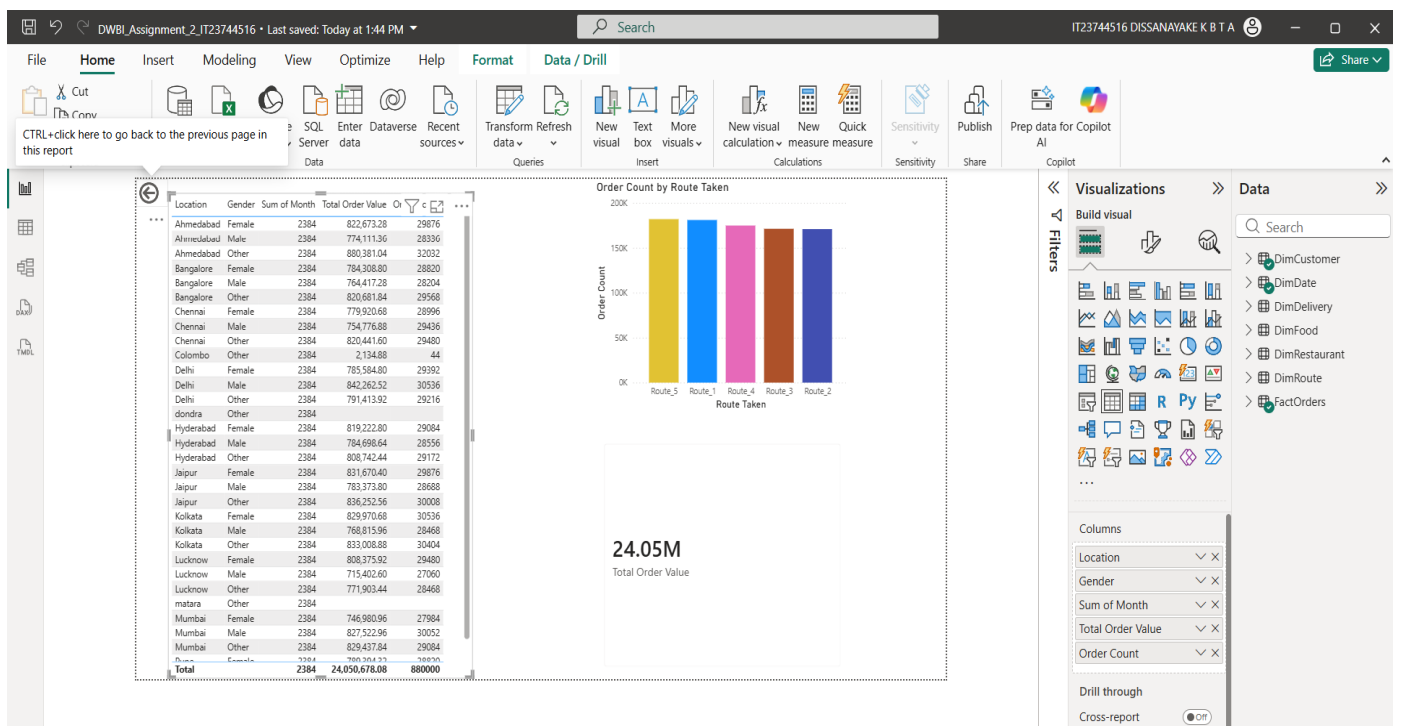
- Drill-through field: Route Taken (or Location)
- Table showing detailed data
- Chart showing breakdown (e.g., Delivery Method)

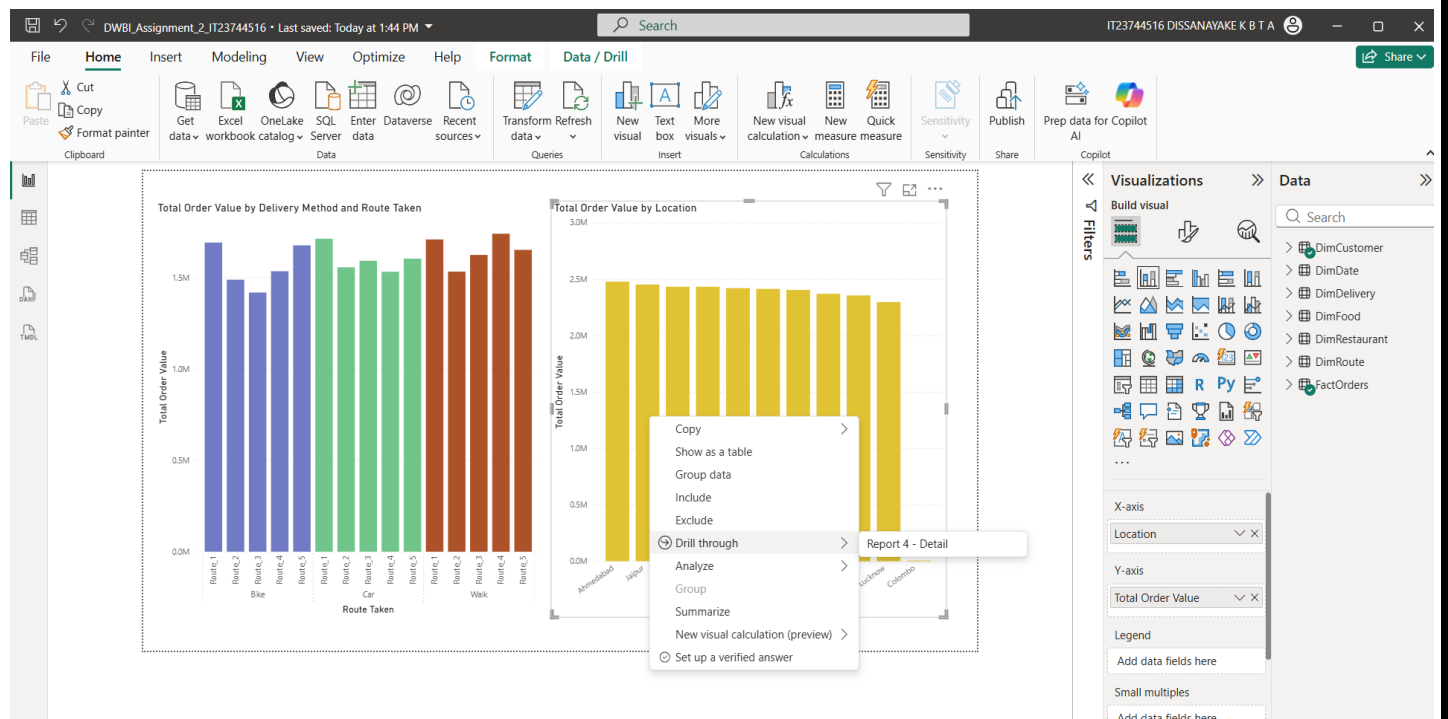
Functionality:

- Right-click on a value → Drill-through → Detail page
- Back button used for navigation

Purpose:

To provide detailed insights for selected categories.



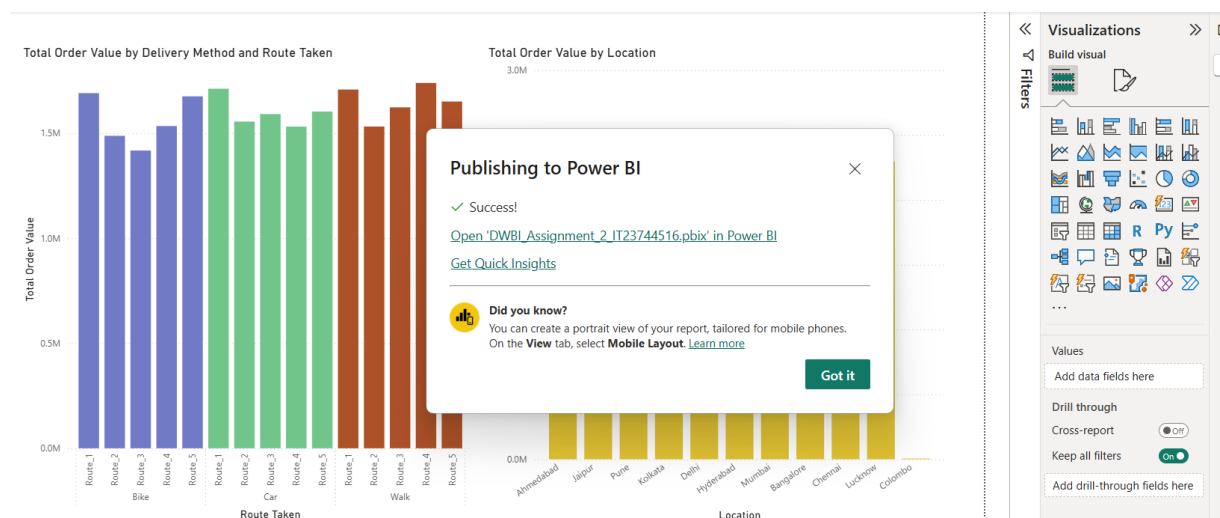


4.6 Publishing to Power BI Service

The report was published using:

- Power BI Desktop → Publish
- Selected “My Workspace”

Link : https://app.powerbi.com/links/OZYQGkhM44?ctid=44e3cf94-19c9-4e32-96c3-14f5bf01391a&pbi_source=linkShare



References:

<https://courseweb.sliit.lk/course/view.php?id=4696> <https://www.geeksforgeeks.org/etl-process-in-https://learn.microsoft.com/en-us/analysis-services/analysis-services-overview?view=sql-analysis-services-2025>
<https://www.microsoft.com/en-us/power-platform/products/power-bi>